Smart E-Waste management system using AI in university Environment

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Abstract:

Electronic waste (e-waste) has become a rapidly growing environmental challenge due to the continuous increase in the consumption and disposal of electronic devices. Improper handling of e-waste leads to environmental pollution, health risks caused by hazardous materials, and the loss of valuable recyclable resources. In many regions, the absence of structured and technology-supported management systems limits effective classification and recycling processes. This graduation project aims to design a structured, technology-based system for managing electronic waste through efficient classification and data-driven workflow design. The project focuses on system analysis, architectural design, and implementation planning to support sustainable recycling practices and improve decision-making in e-waste management.

Abstract (Arabic version)

أصبحت النفايات الإلكترونية تحديًا بيئيًا متناميًا بسرعة نتيجةً للزيادة المستمرة في استهلاك الأجهزة الإلكترونية والتخلص منها. ويؤدي سوء التعامل مع هذه النفايات إلى تلوث البيئة، ومخاطر صحية ناجمة عن المواد الخطرة، وفقدان موارد قيّمة قابلة لإعادة التدوير. وفي العديد من المناطق، يُعيق غياب أنظمة إدارة منظمة ومدعومة بالتكنولوجيا عمليات التصنيف وإعادة التدوير الفعّالة. يهدف مشروع التخرج هذا إلى تصميم نظام منظم قائم على التكنولوجيا لإدارة النفايات الإلكترونية من خلال التصنيف الفعّال وتصميم سير العمل القائم على البيانات. ويركز المشروع على تحليل النظام، والتصميم المعماري، وتخطيط التنفيذ لدعم ممارسات إعادة التدوير المستدامة وتحسين عملية اتخاذ القرارات في إدارة النفايات الإلكترونية.

أصبحت النفايات الإلكترونية اليوم مشكلة خطيرة نظراً للتطور التكنولوجي السريع والواسع النطاق، وتزايد استبدال الأجهزة الإلكترونية بوتيرة أسرع من أي وقت مضى.

يتم استخدام العديد من الأجهزة لفترة وجيزة ثم التخلص منها، حتى وإن كانت لا تزال تعمل أو تحتوي على أجزاء أو مواد مفيدة وقيمة.

تنتشر هذه المشكلة والكارثة في الجامعات والشركات والمؤسسات الكبرى، بل وفي مختلف البيئات. فهي تُلحق الضرر بالبيئة وتؤدي إلى مشاكل صحية خطيرة. لذا، فإن تطوير الذكاء الاصطناعي يُتيح فرصةً واعدةً لحل هذه المشكلة، ورفع كفاءة إعادة التدوير، والاستخدام الأمثل للموارد.

يهدف مشروعنا إلى تحسين وتطوير الإدارة الضعيفة للأجهزة الإلكترونية، ومعالجة نقص التتبع السليم لها، وإيجاد أفضل السبل لتحويل هذه النفايات إلى موارد قيّمة ومربحة، وذلك من خلال تطوير نظام ذكي يُساعد في تحديد وتصنيف النفايات الإلكترونية بكفاءة عالية، بهدف الحد من النفايات ودعم الاستخدام المستدام للموارد.

1. Chapter 1:

1.1 Introduction

Today, electronic waste has become a serios problem because technology is developing very fast and in a large way, and electronic devices are replaced more than before.

Many devices are used for a short time and then thrown away, even though they may still work or contain some useful and valuable parts or materials.

This problem and disaster are common in universities and big companies and institutions, also in all types of environments. in the first place it damages the environment and leads to serious health problems, so on, the opportunity of AI development is capable and helps to solve this problem and Increase efficiency of recycling, and greater utilization of resources.

Our project aims to improve and develop the poor management and the lack of proper tracking of electronic devices and find the worthiest way to make this garbage valuable and profitable, by developing an intelligent system that helps identify and classify electronic waste in a better and the most efficient way, in order to reduce waste and support sustainable use of resources.

1.2 problem statement:

Birzeit university have thousands of electronic devices estimated about 30000 unit, which makes managing and tracking electronic waste hard and difficult without a prober automated system , although barcode records are available for all devices which going to be used as entry for the system and if missing or not classified in database yet the option of image recognition going to be available , so we infer to clear need of a smart e-waste system management that combines barcode data with image base identification to improve tracking , classification and decision making in the university.

1.3 Research questions:

How can AI be used to improve e-waste management in university environment?

How can our system make this electronic waste valuable and can be source of cost reduced and how it will give the best wasting method?

What technology solution can improve e-waste collection tracking and recycling efficiency?

1.4 Project goals:

The main objectives of the project are as follows:

- To promote the reuse of consistent and functional parts of discarded electronic devices in new projects, thereby reducing waste and supporting green environment.
- Develop an AI-powered system that tracks, collects and manages e-waste in Birzeit university using database integration and classification.
- Reduce improper disposal of e-waste in university or environment.
- To reduce financial burden of e-waste management by introducing low-cost solutions.
- To develop and design a system that can be scaled and shifted in other universities or environments.

1.5 E-Waste Situation in Palestine

In Palestine, a lot of old electronic devices and batteries pile up every year. People usually throw them on the streets or send them to landfills without any safe treatment. Some people buy these used devices from homes, fix a few parts, or sell them as scrap metal. According to engineer Hussein Abu Aoun, there is no special system for e-waste in Palestine, so everything ends up in normal landfills. This shows that we really need a clear plan to manage e-waste in a safe and useful way.

Figure 1 public landfill in palestine

The amount of cadmium in just one mobile phone battery is enough to pollute about 600 cubic meters of groundwater. This shows how dangerous even a single battery can be if it is not disposed of properly.

Figure 2 Electronic waste burned to extract minerals in the West Bank

A report done by the Ministry of environment quality authority sorted the amount of toxic materials and water pollution from electronic devices according to west bank cities as shown below in figure 2 and figure 3 .

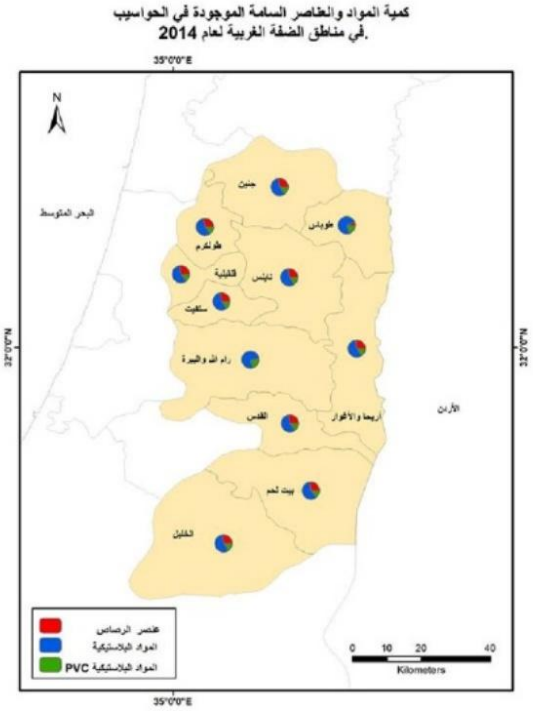

Figure 4 The amount of toxic materials present in computers

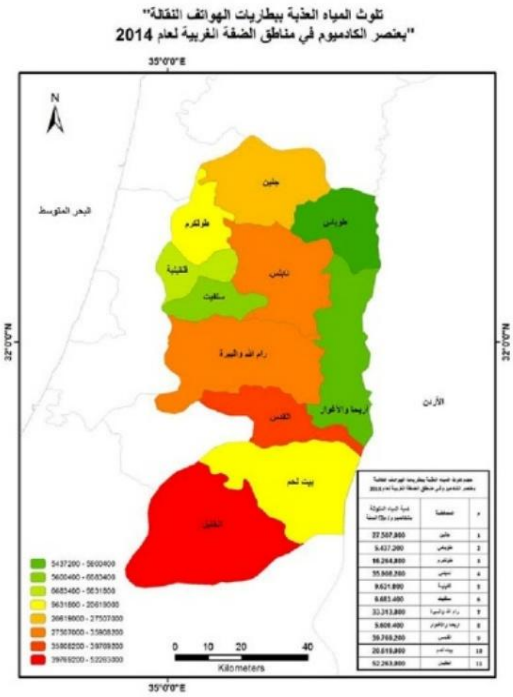

Figure 3 - water pollution in west bank cities

2. Chapter 2: Literature Review

2.1introduction:

Literature review and study previous studies and got important keys from them is essential and important to understand the current state of knowledge on e-waste management. Also, it provides insights into how researchers have approached the problem from different perspectives, including technical prediction methods, reviewing these works helps identify strengths weakness. limitations and gaps that this project aims to address.

2.2 previous studies, research and works

Ali et al. (2023) applied machine learning regression models to predict global e-waste volumes. Their study achieved high accuracy but lacked regional customization, limiting its practical application in local contexts.

Newar & Deka (2025) examined India's evolving legal framework for e-waste management. They highlighted the introduction of Extended Producer Responsibility (EPR) and amendments from 2011 to 2023. Despite strong laws, enforcement remains weak, and the informal sector continues to dominate recycling practices.

Nordin et al. (2025) investigated behavioral determinants of e-waste recycling in Malaysia. Their survey revealed that policy and regulation are the strongest predictors of recycling behavior, followed by moral obligation and knowledge. However, attitude and time availability were not significant factors, and recycling infrastructure remains limited.

Thao et al. (2024) conducted a case study on sustainable e-waste management at Ho Chi Minh City University of Technology. The study analyzed e-waste generation, flow, and environmental impact within a higher education institution. Findings revealed that universities are significant contributors to e-waste due to frequent upgrades of lab equipment and student devices. The study emphasized the lack of structured collection systems and proposed campus-wide policies and awareness programs as solutions.

2.3 comparative analysis

2.3.1 Scope of studies

- India (Newar & Deka, 2025): Legal and policy framework, enforcement gaps.
- Malaysia (Nordin et al., 2025): Behavioral determinants of recycling practices.
- Prediction Models (Ali et al., 2023): Technical forecasting of e-waste volumes.
- Vietnam (Thao et al., 2024): University case study, institutional management.
- Our project: Smart university-based e-waste management system integrating IoT, FSM models, policy, and awareness.

2.3.2 strengths:

- India: Comprehensive legal framework.
- Malaysia: Strong behavioral insights.
- Prediction Models: High forecasting accuracy.
- Vietnam: Practical institutional application.
- our Project: Integrated approach combining technical, legal, and behavioral aspects with real implementation.

2.3.3 Weaknesses:

- India: Weak enforcement, informal sector dominance.
- Malaysia: Limited to one region, infrastructure gaps.
- Prediction Models: Lack of local validation.
- Vietnam: Focused on one university, limited generalizability.
- Our Project: Requires resources and pilot testing; scalability beyond university needs validation.

2.3.4 common insights:

Most studies agree that electronic waste is a growing challenge and that universities need better systems to manage it. Researchers often highlight recycling, awareness programs, and policy frameworks as common solutions. However, a shared limitation across the literature is the lack of artificial intelligence applications to support decision-making. Few attempt to use machine learning to analyze device data or provide transparent recommendations. This gap shows the importance of developing an AI-powered framework, starting at Birzeit University, that can later be adapted to other institutions and languages.

2.3.5 Research gaps

None of the studies fully integrate technical prediction, legal enforcement, behavioral factors, and institutional models into a unified framework

2.4 summary of literature review

Study	Scope	Strength	Weakness	contribution
India (Newar & Deka, 2025)	Technical forecasting of global e-waste	High accuracy in ML models	Lack of local validation	Provides predictive tools but not practical implementation
Malaysia (Nordin et al., 2025)	Legal and policy framework	Comprehensive laws and er framework	Weak enforcement in formal sector dominance	shows policy gaps
Prediction Models (Ali et al., 2023	Behavioral determinants	Strong insights on regulation and moral obligations	Limited to one region, infrastructure gaps	Shows social and awareness impact
Vietnam (Thao et al., 2024):	University case study	Practical institutional application	Focused on one university. limited	Demonstrates campus level solutions
Our project	Smart university-based system	Integrated approach (Iot, FSM, policy awareness)	Needs resources and pilot testing	Fill technical gaps can be applied on another enviroments .

Figure 5 literature revie summary table

2.5 conclusion

Research in the field shows that electronic waste has become a serious global issue, and most universities still lack organized systems to deal with it effectively. Earlier studies emphasize the role of machine learning and data-driven methods in improving waste management, but many of these works are limited by small datasets, narrow focus on recycling, or weak transparency in their results. These shortcomings highlight the need for a more complete solution that can handle large amounts of data, provide clear predictions, and explain decisions in a way that users can trust.

This project aims to fill that gap by applying machine learning to Birzeit University's device records, using explainability tools like SHAP and LIME, and presenting results through a simple dashboard. In this way, the framework builds on past research but offers a more practical and transparent approach that can later be adapted to other institutions.

3. Chapter 3:Methodolgy

The project focuses on data driven modeling and analysis for sustainable e-waste management, it integrates behavioral data, historical records and predictive modeling to design an explainable model for decision making, process begins from collecting detailed information about electronic devices, then trains a machine learning model to analyze their lifecycle and identifies valuable components. And aims to maximize reuse and profitability.

3.1 Current Data Collection Efforts

We have personally visited and engaged with representatives from Birzeit university from Procurement Department and computer center.

data was collected directly from university staff responsible for handling electronic devices. The staff provided institutional records and explained the current practices, which will be formally shared within two days. According to their account, the university lacks a clear and effective policy for managing electronic devices. The process currently follows these steps:

1. attempt to repair the device.
2. If repair fails or the device is outdated, it is sent to storage.
3. Devices often remain in storage for months or years without proper recycling.
4. Eventually, they are sold at auctions at very low prices, treated as scrap metal rather than valuable electronic equipment.

The primary dataset consists of more than 10,000 electronic devices from the university inventory. Each record includes barcode, model, purchase date, and initial cost. This dataset provides a comprehensive view of the institution's electronic asset lifecycle. Which is going to be used also for model learning.

This information highlights the absence of structured e-waste management policies and provides the foundation for designing a sustainable framework in this project.

3.2 data preprocessing

- Data Cleaning: Remove duplicates, fix inconsistencies.
- Normalization: Standardize costs and purchase dates.
- Categorization: Group devices by type and lifecycle stage.

3.3 model architecture:

- Type: Supervised learning (Decision Tree / Random Forest).
- Inputs: Device age, cost, type, maintenance history.
- Outputs: Prediction of valuable components, recommended disposal strategy (reuse, recycling, resale).
- Training: Labeled dataset from university records and expert input

3.4 explainability tools:

SHAP: To identify which features (age, cost, type) most influence predictions.

LIME: To explain individual device recommendations.

Dashboard: Visual interface showing predicted component value, recommended action, and explanation for transparency.

3.5 data split:

In this project, the dataset of more than 10,000 devices will be divided into three parts. 70% of the data will be used for training and learning, so the model can learn the patterns between device features and outcomes. Another 15% will be set for validation to adjust the model and check its accuracy during building. The final 15% will be kept for testing to evaluate how well the model performs on new data it has never seen before. This planned split will help the model stay reliable and avoid problems like overfitting.

3.6 system design:

The system design defines the **architecture and components** of the e-waste management system at Birzeit University. It explains what the system is, how its modules interact, and the technical rationale behind each component.

Core Components:

1. **E-Waste Input and Inventory Module** – Captures detailed information about devices, including type, model, condition, and location. Prepares data for classification.

2. **AI Classification Engine** – Uses machine learning models (CNNs, transformer-based networks) to classify devices by type and recyclability. Incorporates image recognition and sensor data.

3. **Data Management and Tracking Module** – Centralized database stores device data, classification results, and tracks each item's lifecycle
   .
4. **Workflow Module** – Defines procedures for collection, storage, and disposal.
5. **Analytics Module** – Evaluates performance, provides reports, and identifies areas for optimization.

3.7 system flowchart:

The system will begin by gathering information about all electronic devices in the university, including details such as barcode, model, purchase date, and cost. After collecting the data, it will be cleaned and organized to remove mistakes and prepare it for analysis. The next step will be training a machine learning model that studies the devices and predicts which parts have the most value. Based on these predictions, the system will suggest the best way to handle each device, whether by reuse, recycling, or resale. Finally, the results will be explained through tools that show why each decision was made, and everything will be presented in a clear dashboard for easy understanding.

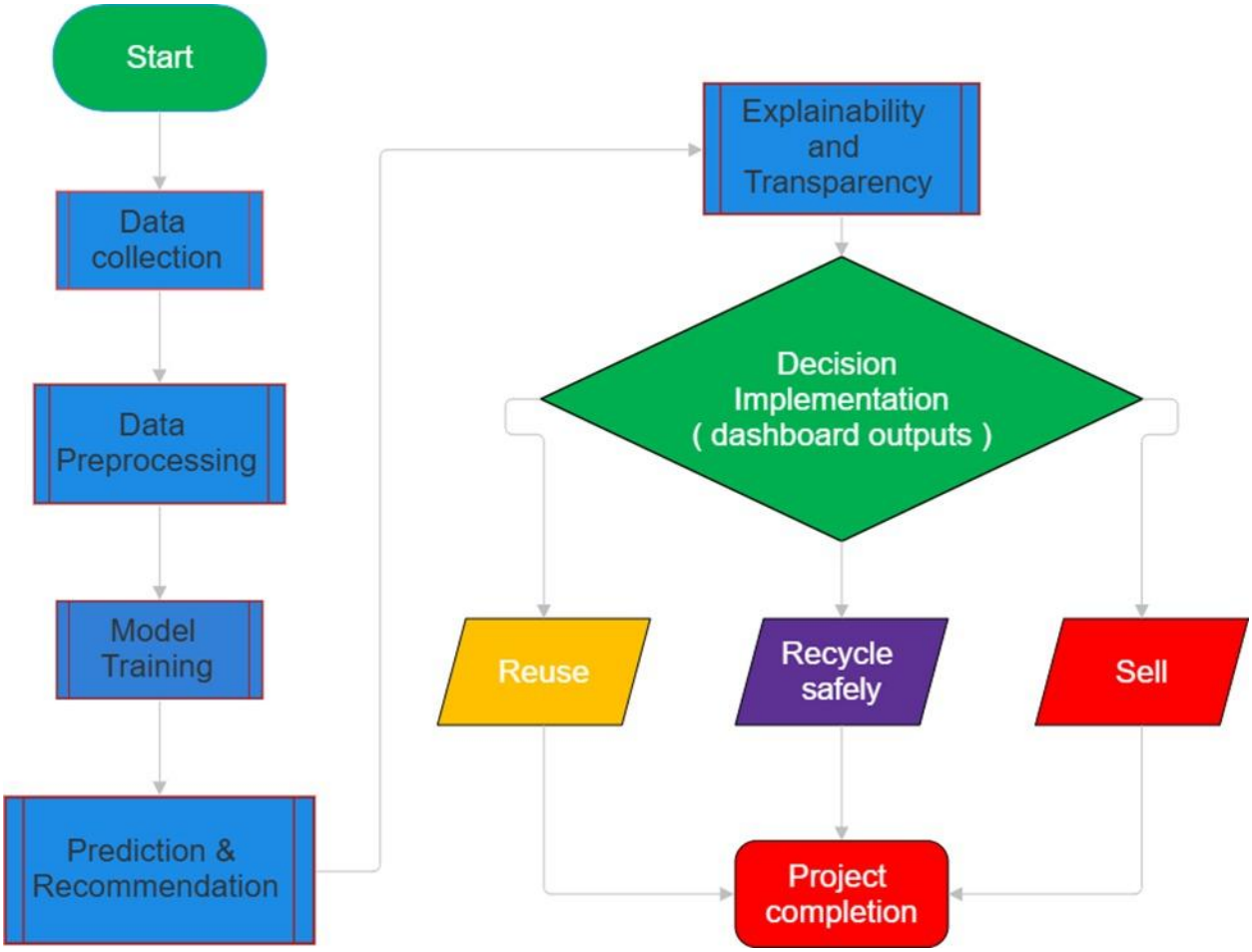

Figure 6 system flow chart diagram

4. Chapter 4: Results and discussion

4.1 expected results and performance

The proposed e-waste management system at Birzeit University is expected to achieve the following outcomes:

1. **Reduced Effort and Time**
    - The system will automatically classify electronic devices, reduce manual work and save significant time in managing e-waste.
2. **Cost and Resource Savings**
    - Efficient organization of collection, classification, and storage will lower operational costs and make better use of available resources.
3. **Green Environment and Sustainability**
    - By increasing recycling rates and reducing improper disposal, the system will contribute to a cleaner campus and a greener environment.
    - Supports the university's sustainability goals and promotes environmentally responsible practices.

4. **Support for Decision-Making and Process Improvement**
    - Provides accurate data on classification, workflow efficiency, cost savings, and environmental impact.
    - Helps administrators optimize operations and make informed, environmentally conscious decisions.

Chapter 5: conclusion and future work:

This system will first be applied at Birzeit University to help manage electronic waste in a smarter and more sustainable way. It will use machine learning to find valuable parts in devices and suggest the best way to reuse, recycle, or resell them. After testing and improving the system locally, it can be adjusted to work in other universities or regions. The dashboard and explanations can also be translated into different languages to make it easier for more people to use and benefit from the system.

References

[1] H. Rantisi, "لا آلية للتخلص من النفايات الالكترونية في فلسطين .. والسبب؟",

بوابة اقتصاد فلسطين, Jul. 31, 2017

Available: https://www.maan-ctr.org/magazine/article/1445/

[2] "AI Optimized Circular Economy Models for E Waste Management In The Tech Industry: A Data Driven Approach,"

Int. J. Environ. Sci., pp. 1104–1112, Aug. 2025. .

Available: https://www.researchgate.net/publication/399753470_AI-driven_circular_economy_optimization_in_waste_management_A_review_of_current_evidence

[3] B. I. Oladapo, M. A. Olawumi, and F. T. Omigbodun,

"AI Driven Circular Economy of Enhancing Sustainability and Efficiency in Industrial Operations,"

Sustainability, vol. 16, no. 23, p. 10358, Nov. 2024.

Available: https://doi.org/10.3390/su162310358

[4] "النفايات الإلكترونية: عجز فلسطيني عن تدويرها",مركز العمل التنموي معاً, Dec. 2020.

Available: https://www.maan-ctr.org/magazine/article/3410/

[5] A. Hamdouni, "The Role of Artificial Intelligence in Enhancing ESG Outcomes: Insights from Saudi Arabia,"

J. Risk Financial Manag., vol. 18, no. 10, p. 572, Oct. 2025.

Available: https://doi.org/10.3390/jrfm18100572

[6] "The Global E-waste Monitor 2024," United Nations University, 2024

Available: https://ewastemonitor.info